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Interpretable Machine Learning for Municipal Residential Gas Consumption: Integrating Climate, Building Stock, and Socioeconomic Drivers

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Abstract

Dutch municipalities lead the phase-out of residential natural gas, and need to know why gas use differs between them. We merge four Dutch open datasets on climate, building stock, and socioeconomic conditions into a panel of 2,905 municipality-years (408 municipalities, 2016 to 2023). A global Random Forest estimated annual gas use per dwelling from 14 predictors, without past consumption, and SHapley Additive exPlanations (SHAP) ranked them. On the held-out 2020 to 2021 test set the Random Forest reached an R^2 of 0.78 (95% CI 0.73 to 0.81) and a mean absolute percentage error of 7.6%. A persistence baseline that repeats each municipality's last known value was more accurate (0.86 and 5.9%), so the model's value is explanation, not prediction. For municipalities left out of training, R^2 fell to 0.62. The strongest predictors were average floor area, heating degree days, population density, and the pre-1975 dwelling share, holding 73.6% of the SHAP importance. Building-stock variables carried most of the accuracy; climate did not improve it. In the 2022 to 2023 energy crisis the predicted level broke down ($R^2 = -5.83$), but the municipal ranking held (Spearman 0.92 in 2022 and 0.85 in 2023). Gas use above the municipal median amounts to 0.48 Mt CO_2-eq per year, linking the results to SDG targets 7.3, 11.3, and 13.2.

Keywords: residential natural gas consumption; municipal energy planning; random forest; explainable artificial intelligence; SHAP; building stock; socioeconomic factors; Sustainable Development Goals; Netherlands

1. Introduction

The Netherlands is committed to phasing out residential natural gas for heating and cooking by 2050 [1]. This is necessary to meet the climate goals that follow from the Paris Climate Agreement. Right now, natural gas is responsible for most of the residential heating in The Netherlands and meeting the climate goals requires new and innovative ways for heating and cooking systems that are powered by renewable energy sources. These technologies, together with building isolation, should reduce energy consumption. The Dutch Climate Agreement of 2019 gives the municipalities most of the responsiblities for this transition [1]. Each municipality can plan how its neighborhoods will move away from gas, and decides where to prioritize insulation programs, heat pump subsidies, and district heating networks. The policymakers of the municipalities depend on reliable estimates of how much gas dwellings currently use, and on a clear understanding of why some

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areas use much more than others. This is also where three Sustainable Development Goals meet at the local level [2]. Gas use per dwelling tells us something about energy efficiency (SDG target 7.3). But it is not a pure efficiency measure, because it also depends on climate, dwelling size, and the heating system. It is decided by the municipal planning capacity named in SDG target 11.3. And it converts directly into the greenhouse gas emissions tracked under SDG target 13.2.

Average annual gas consumption per dwelling in the Netherlands fell from 1,850 m^3 in 2010 to 800 m^3 in 2024, a decrease of approximately 57 percent [3]. Even though, the differences of consumption between the municipalities remain fairly high. Even municipalities of similar size can have a significant gap in their gas use per dwelling. It is not enough for local planning to know the national level of demand. The policymakers also need to know which characteristics, e.g. building age, climate, income, are connected to either high or low consumption. This gives us two important tasks: trying to predict the annual gas consumption per dwelling from municipal characteristics that are observed, and to explain which characteristics are driving the differences between the municipalities. For the first task machine learning is very valuable, because it can model nonlinear relations and interactions between building data, weather, and socioeconomic conditions. For the second task, interpretability methods such as SHAP [4] can be used, because accurate predictions by itself do not help the policymakers to decide which factors are important.

Machine learning models for energy consumption forecasting have been studied a lot at the building scale, with the availability of building sensor data, and they have consistently found that building characteristics are the strongest predictors on annual energy use [5–7]. The studies at the neighborhood scale look similar, but found that the past consumption is usually the strongest predictor when it is included [8,9]. At the municipal scale, the scale at which the Dutch heat transition is planned, there are only a few studies. Those studies each only cover a part of the problem [10–12]. Gaweł and Paliński [12] are the closest, who predicted annual household gas consumption for Polish municipalities and showed that global models outperform local ones. However, their feature set was limited, with only lagged consumption and population. It did not include climate, or building-stock variables, and no interpretability analysis. In Dutch planning practice, tools estimate the heat demand of dwellings from building characteristics [13,14], but they do not test how well open data explain observed differences between municipalities. Section 2 reviews this literature in detail.

This paper addresses that gap. We merge four Dutch open datasets, which covers climate, building stock, and socioeconomic conditions into a single municipality-year panel. We then use it to predict average annual residential gas consumption per dwelling. We train a global Random Forest on all municipalities together and compare it against a Ridge regression, a mean baseline, and a persistence baseline. The tree-based ensembles are a good fit for this problem, because the panel is small and tabular [15,16]. Then we interpret the model with SHAP to rank and describe the predictors.

The main contributions of this paper are:

- **A multi-source municipality-year panel.** We merge four Dutch open datasets into a panel of 2,905 observations covering 408 municipalities from 2016 to 2023, providing a transparent basis for future benchmarking.
- **An annual municipal model combining climate, building-stock, and socioeconomic data.** We estimate annual gas use per dwelling across municipalities from open climate (KNMI), building-stock (CBS 82550NED), and socioeconomic (CBS Wijk- en Buurtkaart) data, and compare feature groups to test which of them carry the model's accuracy.

- **A SHAP explanation of inter-municipal differences.** We rank and describe the predictors of residential gas use across municipalities. To our knowledge, this is the first SHAP-based explanation of annual residential gas consumption differences at the municipal scale.
- **Out-of-distribution evaluation under an energy price shock.** We evaluate the model on the 2022 energy crisis, and find that level accuracy drops but the ranking of municipalities holds.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 describes the data and methods, and Section 4 presents the results. Sections 5 and 6 discuss the findings and conclude.

2. Literature Review

Machine learning is now often used for forecasting natural gas and residential energy. It's because the relationships between weather, building data, and socioeconomic conditions are often nonlinear [17–19]. The reviews that have been conducted all show that the best model depends on the spatial scale, the size of the dataset and the type of forecasting [17,18]. The short-term forecasts depend mostly on the temperature and the recent consumption. Annual forecasts depend more on slower moving variables, like population, income, urbanization, and the building data [17,18]. This difference is important for municipal planning. The goal is not only to predict the demand, but also to explain why some municipalities use more gas than others.

Most of the recent residential energy studies are done at the building level. These studies usually have detailed sensor data, but they cover a small number of buildings, or some even just one single building. They all agree that building characteristics are important for annual energy use. Olu-Ajayi et al. [5] found that floor area was the strongest predictor for annual residential energy in UK buildings. Vaisi et al. [6] found that building size, floor area, building age, insulation, and heating systems were the strongest predictors of heating energy use. Together, these studies show that the physical building variables are important indicators that any annual energy model needs to perform.

At the other end of the scale are the national studies. They try to predict energy use for a whole country and they use variables such as population size, economic growth, and how urban a country is. Tree-based models perform well here. Farías et al. [20] reached an R^2 of 0.96 for residential electricity in Brazil. Mohammadiziazi and Bilec [21] found that ensemble methods perform better than linear regression in the United States. Deep learning is less common, because datasets are usually too small to train large neural networks. Kernel methods are a strong alternative. Singh et al. [22] forecast annual natural gas use in the United States with a support vector machine and reached a MAPE of 2.7 percent. Gao and Shao [23] reported a MAPE of 0.51 percent for annual gas use in China. For annual forecasts, socioeconomic variables such as GDP per capita, population, and urbanization are the strongest predictors [22,23]. For monthly forecasts climate variables such as heating degree days matter more, because they follow the seasons [24]. Cui et al. [7] applied SHAP to U.S. household survey data and found that floor area, heating fuel, climate, and building age were the strongest predictors. However, the problem for local policy is that national models are looking at country totals, meaning they hide the differences between places. They cannot explain why one municipality uses more gas than another.

Closer to the municipal problem are the neighborhood-scale studies. Jafarigorzin et al. [8] predicted annual household energy use for postal code areas with building, demographic, and climate variables. They found that lagged consumption was the strongest predictor when it was included. When they removed it, the number of households, surface area, and construction year became the strongest predictors. Colverd et al. [9] predicted

domestic gas and electricity use for small neighborhoods in England and Wales. They found that building count and floor area were the strongest predictors, and that temperature followed closely. The studies show that lagged consumption can make predictions very accurate, but it also hides the structural variables that matter for retrofit and heat transition planning. So, a model that mostly repeats last year's value cannot explain why one area uses more gas than another, or how that pattern might change under a potential new policy.

Only a few studies have researched at the municipal scale. Panek and Włodek [10] forecast daily natural gas demand for a Polish city. They found that Random Forest gave the best accuracy, and temperature and lagged consumption were the strongest predictors. This is useful for daily operations, but it again depends on a short-term signal. Mittakola et al. [11] studied 1,000 U.S. counties and used SHAP to explain why counties differ. They found that building age, household size, employment, and income were the strongest predictors. Their study is relevant because it combines climate, building, and socioeconomic variables in one model. However, their target is temperature sensitivity, not annual consumption. Gaweł and Paliński [12] forecast long-term household natural gas consumption for Polish municipalities. They found that global models, trained on all municipalities together, performed better than local models. Their finding supports the use of one global model when the time series of each municipality is short. The main limit is that they use past consumption, population, and dwelling counts. They did not include climate variables, of building-stock characteristics. And it gives no SHAP-based explanation of why the municipalities differ.

Tree-based ensembles are a strong starting point for tabular energy prediction. Random Forest [25], XGBoost [26], LightGBM, and CatBoost often perform well, because they can model nonlinear relations and interactions without needing large training sets [6,15,20]. Deep learning works well for large hourly datasets, but an annual municipal panel is much smaller and has structured tabular variables instead of sensor sequences [16,27]. The same pattern runs through the studies discussed above. Tree-based models were the best performing family at the building scale [6,15], the neighborhood scale [8,9,28], the national scale [7,21,24], and the municipal scale [10,11]. This consistency across four spatial scales supports the use of a tree-based method as a baseline for annual municipal prediction.

Municipal energy planning needs more than accurate predictions, it needs interpretability as well. The policymakers need to know which variables influence high or low gas use. SHAP [4] has already been used in energy studies at the building, national, and neighborhood scale [7,11,24,28]. These studies show that SHAP can make tree-based energy models easier to understand. To our knowledge, SHAP has not yet been used to explain differences in annual residential gas consumption between municipalities.

Dutch research and planning tools give some useful background here. Brounen et al. [29] studied more than 300,000 Dutch homes and found that gas use depends mostly on the dwelling itself, such as its construction period and type. Electricity use, in contrast, follows household composition more closely. Majcen et al. [30] compared actual gas use with the theoretical use implied by energy labels and found large gaps: dwellings with poor labels used much less gas than predicted. Municipalities also have tools they can use in planning practice. One is the PBL Startanalyse [14], which uses the Vesta MAIS simulation model to compare the costs of natural-gas-free heating strategies per neighbourhood. Another is the Referentieverbruik method [13,31]. It estimates the heat demand of each dwelling with a linear regression on measured 2020 gas use in CBS microdata. The predictors are dwelling type, construction period, energy label and floor area, and the results are published as one file per municipality. But neither tests how well open municipal data explain the observed differences in annual gas use between municipalities.

The gap in the literature is more specific than a missing set of features. We found only three machine learning studies at the municipal scale [10–12], and just one of them targets annual gas consumption [12]. SHAP appears in a single municipal study [11], where it explains how strongly each county responds to temperature. How much gas a county uses over a year is left unexplained. Dutch planning tools estimate demand bottom-up and treat whatever local differences remain as a correction factor [13]. But no study so far estimates observed annual residential gas use per dwelling for all municipalities from open data. Such a model should also be compared with simple baselines, including persistence, and SHAP can then show which characteristics explain the differences between municipalities. This paper does that for the Netherlands.

3. Materials and Methods

3.1. Study Design and Prediction Task

This study uses a supervised regression design to predict average annual residential natural gas consumption at the municipal level in the Netherlands. The unit of analysis is the municipality-year: each observation covers one municipality in one calendar year, paired with a target value and a set of explanatory variables covering climate, building stock, and socioeconomic context.

The target variable is average annual gas consumption per dwelling, measured in cubic meters per dwelling per year, taken from CBS 81528NED and denoted `gas_m3` in the processed dataset. Because the target is expressed per dwelling, municipalities of different sizes end up on the same scale, which makes them directly comparable.

The main task is to estimate `gas_m3` for municipality-year observations with a single global model trained on all municipalities at once, rather than a separate model per municipality. It learns patterns that hold across the national panel. This follows from the gap identified in Section 2: no study so far estimates observed annual gas use for all municipalities from open data and uses SHAP to explain the differences between them.

The model estimates consumption for a year that is already over. It does this after the conditions of that year are known, so it works as a retrospective annual estimate. It cannot predict consumption in advance.

The reason is in the input data. The climate features use observed temperatures for the target year: winter temperature covers October of year $y-1$ to April of year y, and HDD covers the full calendar year y. So the winter value is only complete when the heating season ends, and HDD only at the end of the year. The building-stock and socioeconomic features have the same limit. They are CBS values for year y, and CBS publishes them after that year has finished.

The point of the modelling step here isn't to introduce a new algorithm. A well-established tree-based model is used as an empirical tool, for two reasons: it tests how well the combined feature set predicts municipal gas consumption, and it helps explain which variables drive the differences between municipalities. Predictive performance gets checked against simple baselines, but the real interpretation rests on SHAP values from the Random Forest model.

3.2. Data Sources

This study draws on four Dutch open datasets, summarized in Table 1.

Table 1. Source datasets used in this study. Years are the published editions downloaded; the analysis uses 2016 to 2023 (Section 3.3).

Dataset	Publ.	Years	Municipalities	Variables
CBS 81528NED [3]	CBS	2010 to 2024	463	Target: `gas_m3`; auxiliary: district heating share
CBS 82550NED [32]	CBS	2012 to 2026	441	Floor area; pre-1975 share; single-family share
Wijk- en Buurtkaart [33]	CBS	2013 to 2024	440	9 socioeconomic variables
KNMI daggegevens [34]	KNMI	2010 to 2024	345	HDD; avg. winter temperature

The municipality counts in Table 1 refer to the filtered source datasets before final merging. The final merged panel contains 408 municipalities, while KNMI climate values are imputed for boundary-change municipalities.

3.2.1. CBS 81528NED

CBS 81528NED (*Energieverbruik particuliere woningen; woningtype en regio's*) comes from Statistics Netherlands (CBS), published through the StatLine open data portal [3]. It gives average annual energy use per dwelling, split by housing type and region, for 2010 to 2024. The target variable here is `GemiddeldAardgasverbruik_1`, renamed `gas_m3` and measured in cubic meters per dwelling per year (m^3/dwelling/year). Using consumption per dwelling rather than a municipal total corrects for size, so a big city and a small village become directly comparable.

The raw dataset has 66,480 rows across eight columns. Its region column mixes six spatial levels together: 463 municipalities, 40 energy-transition regions, 30 energy-statistics regions, 13 provinces, 5 macro-regions, and the national total. The housing type column has eight categories — detached houses, apartments, owner-occupied dwellings, rental dwellings, and a total across all types, among others. For this study, only municipality-level rows and the total housing category (*Totaal woningen*) were kept, so each row ends up representing one municipality in one year. Two non-geographic entries, *Buitenland* and *Gemeenten; niet in te delen*, were dropped as well. What's left is 6,945 rows covering 463 municipalities over 15 years.

The dataset also includes a district heating share column, recording the fraction of dwellings connected to a district heating network. CBS reports this because it matters for interpreting gas use. Dwellings on district heating often show very low or zero gas consumption. But this table doesn't include the actual delivered heat volume [3]. The processed panel keeps this column as an auxiliary variable, useful for making sense of municipalities with unusually low gas numbers. How it's handled is covered in Section 3.3.

About 17.7% of `gas_m3` values are missing in the filtered subset, and most of that comes down to Dutch municipal boundary reforms (*gemeentelijke herindelingen*). CBS calculates its statistics for whatever municipal boundaries were in force at the time, so once a municipality is dissolved in a reform, it stops appearing in later years. Mergers can break up a time series the same way [35]. Section 3.3 covers how these gaps get handled in the merging step.

3.2.2. CBS 82550NED

CBS 82550NED (*Voorraad woningen; gemiddeld oppervlak; woningtype, bouwjaarklasse, regio*) comes from CBS's StatLine open data portal [32]. It reports housing stock counts and average usable floor area per dwelling, broken down by housing type, construction period, and region, recorded annually as of January 1st for 2012 through 2026. All three building

features used in this study come from this one dataset: average floor area, pre-1975 housing share, and single-family housing share.

The raw file has 338,850 rows across seven columns. As with CBS 81528NED, the region column mixes five spatial levels — municipalities, COROP regions, provinces, macro-regions, and the national total — so after filtering down to municipality-level rows, 441 municipalities remain. Each municipality-year combination breaks down into three housing type categories (single-family, multi-family, total) and 15 construction period values: 12 dated bands running from `1000 tot 1850` to `2015 tot 2025`, plus a post-2025 class (`Vanaf 2025`), an unknown class (`Onbekend`), and a total row.

1975 is the key threshold here. The Netherlands brought in its first mandatory building insulation rules after the 1973 oil crisis [36], so dwellings built before that year went up with no such requirements in place. This dataset gives us what we need to derive all three building features. In 2022, the pre-1975 share ranged from 0% to 74.5% across municipalities, with a median of 48.1%. Average floor area ranged from 76 to 212 m^2, median 131 m^2. Section 3.4 gives the derivation formulas for all three features.

Missing values hit about 22.9% of dwelling count rows and 34.5% of floor area rows. The same boundary reform pattern described earlier explains most of this, on top of CBS suppressing small cells in the floor area column [32]. None of these raw-source gaps carry over into the final 2016–2023 panel, though, since the panel merge only uses active municipality-year combinations. By 2022, 345 municipalities have complete data across all three derived building features.

3.2.3. CBS Wijk- en Buurtkaart

The CBS Wijk- en Buurtkaart (*Kerncijfers wijken en buurten*) is published by CBS as a series of annual open data tables, one per year, covering socioeconomic and demographic variables per municipality, neighborhood, and district [33]. This study uses the 2013 to 2024 editions: twelve tables in total. Unlike CBS 81528NED and CBS 82550NED, which come as single multi-year tables, the Wijk- en Buurtkaart has to be downloaded and concatenated one year at a time. The raw concatenated panel ends up with 201,857 rows and 548 columns. That column count is inflated mainly because CBS renumbers its column keys across editions, so columns from different years need to be matched by name before they can be combined.

After filtering to municipality-level rows, the working subset covers 440 unique municipalities across 12 years (4,453 rows). The municipality count drops from 408 in 2013 to 342 in 2024, the same boundary reform pattern described earlier.

Nine socioeconomic features come from this source: average household size, population density, share of residents aged 65 or older, share of single-person households, average income per resident, share of low-income households, average WOZ property value, share of owner-occupied dwellings, and share of social housing dwellings. Each was chosen for three reasons: a plausible link to residential gas use, at least 70% coverage over the 2016 to 2023 panel, and no overlap with the building features already drawn from CBS 82550NED.

The reasoning behind each variable differs. Average household size can indicate how heating demand is split across the occupants of a dwelling. Income per resident and WOZ value work as rough proxies for whether a household can afford insulation or a more efficient heating system. Elderly residents tend to spend more time at home, which is part of why their share is included here; it may capture something about indoor temperature preferences too. Owner-occupied and social housing shares matter for a different reason: owner-occupiers, private tenants, and social housing providers all face different incentives, and different money, when it comes to renovation.

Gas and electricity consumption are both present in the Wijk- en Buurtkaart, but excluded as predictors since they would leak into the target variable.

Three of the nine features have missing values. Income per resident is nearly complete, just three missing rows in 2024. Low-income share is missing entirely from the 2013 and 2024 editions. WOZ property value wasn't collected until after 2015, so it's absent from 2013 to 2015. The other six features are complete. These gaps are why the final panel is restricted to 2016 to 2023, as covered in Section 3.3.

3.2.4. KNMI

The KNMI dataset comes from the Royal Netherlands Meteorological Institute (KNMI), through its daily climate data endpoint at `daggegevens.knmi.nl` [34]. It holds daily mean temperatures (`TG`) from 51 weather stations across the Netherlands for 2010 to 2024. The raw data has 259,173 rows, and 49 of the stations report at least some values. About 28.4% of the daily temperatures are missing. Most of these gaps come from stations that were not running for part of the period. We leave out the 14 stations with more than 5% missing days, which still leaves 35 stations with dense coverage across the country for interpolation. Station-years with fewer than 330 valid daily readings are also dropped before we calculate any annual features.

From this daily data we build two yearly climate features that reflect heating demand: heating degree days (HDD) [37] and average winter temperature. Section 3.4 gives the exact definitions and formulas. To get a value for each municipality, we apply inverse distance weighting (IDW) using municipality centroids from the PDOK WFS service [38]. The weighting scheme is described in Section 3.3.

The final climate dataset has 345 municipalities over 15 years, so 5,175 cells, and 98.3% of them are complete. The 89 cells with no valid HDD value are mostly island municipalities, plus a few remote locations.

Our four datasets cover yearly climate conditions, the building stock and the socioeconomic context, all from open sources. But having the data side by side only gets us part of the way. Once it is joined into a single municipality-year panel, we can estimate how much these dimensions explain together about the differences in gas use between municipalities, and then use SHAP analysis to find the share of each one.

3.3. Data Preprocessing and Panel Construction

Figure 1 summarizes the full preprocessing and merging workflow. All four datasets are joined on CBS municipality code (GM code) and calendar year. CBS 81528NED and CBS 82550NED identify municipalities by Dutch text name, while the Wijk- en Buurtkaart and KNMI use the GM code (for example, GM0363). A lookup table is built from the Wijk- en Buurtkaart panel, which contains both the Dutch name and the GM code in every annual edition, and maps each municipality name to its GM code for each year. Before applying this lookup, nine municipality names in the two name-based CBS datasets include a provincial label in brackets (for example, "Groningen (gemeente)" and "Beek (L.)") that the Wijk- en Buurtkaart does not use. These names are changed to their plain forms in both CBS 81528NED and CBS 82550NED. Non-municipal rows are removed from CBS 81528NED by excluding regional suffixes for provinces, COROP regions, energy regions, and macro-regions.

The panel runs from 2016 to 2023. Two coverage gaps in the Wijk- en Buurtkaart decide this window. The WOZ property value is missing from the 2013 to 2015 editions, and the low-income household share is missing from the 2013 and 2024 editions. With 2016 to 2023, none of the nine socioeconomic features has a missing year. Adding 2013 to

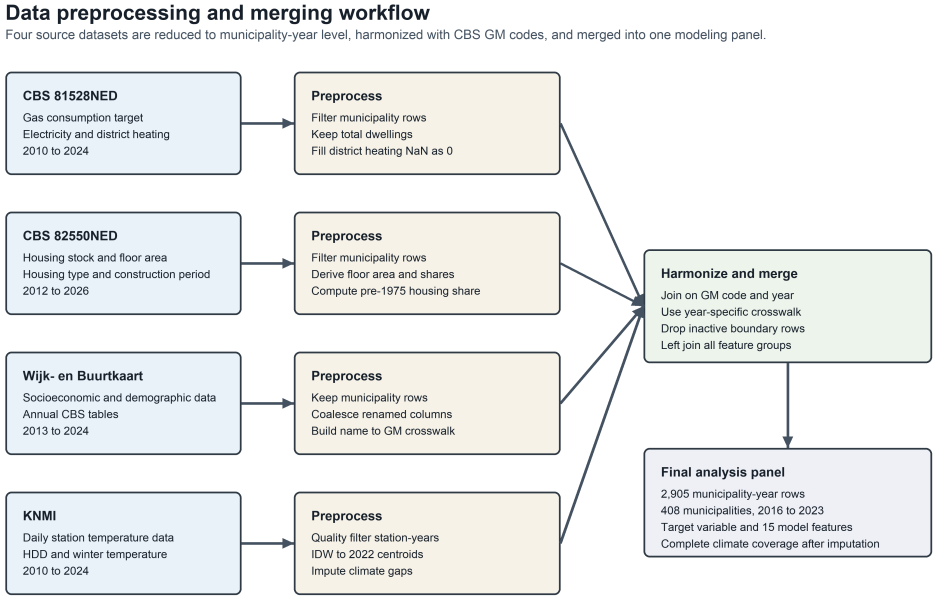


Figure 1. Data preprocessing and merging workflow. The four source datasets are reduced to municipality-year level, harmonized using CBS municipality codes and calendar year, and merged into a panel of 2,905 observations covering 408 municipalities from 2016 to 2023. Climate gaps are imputed after the KNMI join.

2015 would mean imputing the WOZ value for every municipality in those years, so this predictor would carry no observed information there.

Before merging, we reduce each dataset to one row per municipality-year. For CBS 81528NED, we keep only municipality-level rows and the "Totaal woningen" housing type, which groups all dwelling types together. CBS publishes the district-heating share only where ten or more inhabited dwellings are connected [3]. An empty cell therefore means fewer than ten connected dwellings, and we set it to zero. The share is missing for every municipality in 2018, which is a publication gap, so for that year we interpolate each municipality's value from the surrounding years. CBS 82550NED gets the same filtering, and we derive three building features from its housing type and construction period subsets for each municipality-year (see Section 3.4). The Wijk- en Buurtkaart is harder to work with. Its panel has 548 columns, because CBS renames variables from one annual edition to the next. To get the nine socioeconomic features, we match each variable to its column name in every edition, then combine the values into one column and keep the first available value per row. Two features need one more step. Between the 2014 and 2015 editions, the elderly share and the single-person household share switched from percentages to counts, so we divide the count values by total population or total households before using them.

The four datasets are joined one after another with left joins on municipality code and year, starting from CBS 81528NED. A lookup table first gives GM codes to the datasets that use municipality names. Rows in CBS 81528NED that still have no match after this are municipality-years that were not active because of boundary changes: some municipalities had already merged, and others did not exist yet in that year. These rows get no GM code, and we drop them before the first join. That leaves a base of 2,905 rows for 408 municipalities. The joins with CBS 82550NED and the Wijk- en Buurtkaart add columns but remove no rows, since every municipality-year in the base has a match in both sources for 2016 to 2023. The KNMI join is where things go wrong. It leaves 254 municipality-year

cells with no climate values, because the IDW interpolation uses the 2022 municipality centroid file, and that file does not cover every boundary version between 2016 and 2023.

We fill the missing KNMI values in two steps. The first step uses the CBS 2022 boundary file. If a municipality with missing values has a centroid in that file, we find the nearest municipality with valid KNMI data in the same year and copy its climate values. Nearest here means the shortest distance between centroids. This fills 46 rows, all in Zeeland and West Brabant. In that region too few nearby stations reported in 2016 and 2017, and for three coastal municipalities this was true in every year.

But 208 rows are still empty after this step. They belong to municipalities that are not in the 2022 boundary file, because they merged before 2022 or were formed in 2023. Most of them are small municipalities in 2016 to 2018. These rows get the mean for that year across all municipalities with valid climate values. We calculate this mean after the nearest-neighbor step. Every imputed row is flagged in the processed dataset. After both steps, all 2,905 municipality-year cells have climate values.

The final merged panel has 2,905 rows. It covers 408 municipalities over 8 years, from 2016 to 2023. The target variable is `gas_m3`, and the model uses 14 predictors: two climate features and three building features, plus the nine socioeconomic features. We also keep `year`, `electricity_kwh`, `district_heating_pct`, and `total_dwellings` in the processed panel for reference. Municipality names and GM codes are metadata and are not model inputs. The same goes for `climate_imputed`.

3.4. Feature Engineering

Feature engineering turns the four source datasets into one shared set of predictors for each municipality and year. The final feature set has three groups, and each one follows from the research gap in Section 2. Climate variables describe how much heating a year needs. The building variables cover the size and type of the dwelling stock, and also how old it is. On the socioeconomic side, the features describe household composition, income, housing tenure and urban form.

Two yearly climate features come from KNMI daily mean temperature observations. The first is heating degree days (HDD), which measure how much heating is needed in a year. For each valid station s and year y, HDD is computed as

$$HDD_{s,y} = \sum_{d \in y} \max(0, 18 - T_{s,d}), \tag{1}$$

where $T_{s,d}$ is the daily mean temperature in degrees Celsius and 18 °C is the base temperature. The calculation starts at station level. Daily observations are first turned into one value per station and year, and only then are these values assigned to municipalities, using inverse distance weighting from the five nearest valid KNMI stations. Working in this order means missing daily station observations never have to be filled in at municipality level.

The second climate feature is average winter temperature. It is the mean daily temperature over the heating season, from October of year $y-1$ to April of year y, and it is labelled as year y. This gives a more direct measure of how severe a winter was. Both features come from the same temperature series, so some overlap is expected. But their Pearson correlation in the panel is only -0.66 ($r^2 \approx 0.43$), which means average winter temperature picks up variation that cumulative HDD misses. A winter with a few very cold days and a winter that stays mild for months can look quite different on the two measures.

CBS 82550NED provides three building-stock features. Average floor area, `floor_area_m2`, comes from the CBS row where both housing type and construction

period are `Totaal`. It is the average usable floor area per dwelling in a municipality-year. The share of single-family homes is computed as

$$sfh_share_{m,y} = \frac{single_family_dwellings_{m,y}}{total_dwellings_{m,y}}. \quad (2)$$

This share shows how dwelling types are divided within each municipality. Single-family homes are usually larger than apartments and have more wall and roof surface exposed to the outside, so they tend to need more heating. The pre-1975 housing share is computed as

$$pre1975_share_{m,y} = \frac{pre1975_dwellings_{m,y}}{total_dwellings_{m,y}}. \quad (3)$$

The numerator adds up all dwellings in the construction period classes before 1975: `1000 tot 1850`, `1850 tot 1905`, `1905 tot 1925`, `1925 tot 1945`, `1945 tot 1955`, `1955 tot 1965`, and `1965 tot 1975`. These homes were built before modern Dutch insulation requirements, so the share is used as a proxy for insulation quality. It is an indirect measure, since the data say nothing about renovations carried out later.

The nine socioeconomic features come from the Wijk- en Buurtkaart, shortened to WBK in Table 2. CBS changed several column keys between yearly editions, so different names for the same variable are combined before merging. Once names and units are consistent, most variables can be used as they are.

Two variables need an extra unit correction. Elderly share and single-person household share appear as counts in some early editions and as percentages in later ones. The counts are divided by total population or total households before use. Percentage variables from the Wijk- en Buurtkaart stay in percentage points, while `sfh_share` and `pre1975_share` from CBS 82550NED stay as fractions.

Table 2 lists the 14 predictors used in the model. The processed dataset also keeps year, electricity consumption, district heating share, total dwellings, municipality name, municipality code and the climate imputation flag.

Table 2. Model features used for annual municipal gas consumption prediction.

Feature	Source	Unit	Description
`hdd`	KNMI	degree-days	Annual heating degree days, base 18 °C
`avg_winter_temp`	KNMI	°C	Mean temperature from October to April heating season
`floor_area_m2`	CBS 82550NED	m^2	Average usable floor area per dwelling
`sfh_share`	CBS 82550NED	fraction	Share of dwellings that are single-family homes
`pre1975_share`	CBS 82550NED	fraction	Share of dwellings built before 1975
`avg_household_size`	WBK	persons	Average number of persons per household
`population_density`	WBK	persons/km^2	Population density of the municipality
`elderly_share`	WBK	percentage points	Share of residents aged 65 years or older
`single_person_share`	WBK	percentage points	Share of households with one person
`income_per_resident`	WBK	thousand EUR	Average income per resident
`low_income_share`	WBK	percentage points	Share of households with low income
`woz_value`	WBK	thousand EUR	Average WOZ property value
`owner_occupied_share`	WBK	percentage points	Share of dwellings that are owner occupied
`social_housing_share`	WBK	percentage points	Share of dwellings owned by housing associations

Note: `sfh_share` and `pre1975_share` are expressed as fractions (0–1); WBK percentage features are expressed as percentage points (0–100).

3.5. Predictive Modeling Approach

The predictive modelling step tests whether the combined climate, building-stock and socioeconomic features can estimate annual municipal gas consumption. The model has two jobs here. It measures how much predictive value the integrated feature set has, and it provides the basis for the SHAP interpretation in Section 3.7. Comparing many machine learning algorithms or building a complex forecasting system is outside the scope of this paper, so the setup is kept small and easy to reproduce, with every choice stated openly.

The reference Random Forest and Ridge models predict `gas_m3`, the average annual residential natural gas use per dwelling, from the same 14 predictors (Table 2). Electricity consumption is left out, because it is a closely related energy outcome and should not be used to explain gas use. Municipality identifiers are also left out. Without them, the model has to learn from observable municipal characteristics and cannot rely on a label for each municipality.

We ran two checks to see how sensitive our results are to district heating. In the first, we added the share of district heating as an extra predictor. In the second, we refitted the Random Forest with an adjusted target: gas consumption divided by the share of dwellings that are not connected to district heating.

The eight-year panel is split by calendar year into three windows, each with its own role. The training set covers 2016 to 2019 (1,513 observations). The in-regime test set covers 2020 to 2021 (707 observations) and checks whether relationships learned from earlier years still predict later observations for known municipalities under similar market conditions. The years 2022 and 2023 (685 observations) are held out as a separate out-of-distribution (OOD) stress-test set. They are never used for training or model selection. These are the years of the 2022 European energy crisis, when high prices and public calls to save energy

came with a fall in consumption far below what climate and building characteristics alone would predict. Keeping these years apart makes it possible to test a clear hypothesis. A model built on structural municipal features should capture the differences in consumption between municipalities. But it should fail to predict a national demand shock that comes from outside the feature set. Because the splits follow calendar years, the windows do not overlap in time. Most municipalities do appear in more than one window, though, so the design does not test prediction for completely unseen municipalities.

The main performance evaluation (Section 4.2) and the SHAP interpretation (Section 3.7) use only the training and in-regime test sets, which describe the model under the conditions it was fitted on. The OOD results are reported on their own in Section 4.4, where they show how much accuracy drops once consumption leaves the training regime. Since the crisis years play no part in fitting or interpretation, the feature-importance results describe structural relationships and are not mixed up with short-term crisis behavior. The descriptive analysis in Section 4.1 still uses the full 2016 to 2023 panel.

Four models are estimated. The first is a mean baseline. It predicts the 2016 to 2019 training mean of `gas_m3` for every observation in the 2020 to 2021 test set, uses no explanatory variables, and sets a floor that any useful model should beat. The second is a persistence baseline, which predicts each municipality's 2019 value for both 2020 and 2021. It has no predictors, only the municipality's own past gas use. It uses no data from after 2019, so it follows the same temporal cutoff as the other models. Its inputs differ, however: persistence uses only the municipality's own past gas use, whereas Ridge regression and the Random Forest use the 14 predictors and no past consumption. The third model is Ridge regression on the same 14 predictors. It is a transparent benchmark for a simple additive relationship between the predictors and gas use, and its L2 penalty limits the effect of correlated predictors. The predictors are standardized before fitting, because the penalty only treats them equally when they are on a similar scale. Otherwise some predictors would be penalized more than others just because of their units. The regularization strength is chosen from 13 log-spaced values between 10^{-3} and 10^{3} by efficient leave-one-out cross-validation on the training set (the `RidgeCV` default), which minimizes squared error and involves no shuffling or random seed. Leave-one-out lets other years of the same municipality stay in the fitting data, so we repeated the selection with the temporal folds described below. Several predictors are correlated, so the Ridge coefficients are not interpreted and the model serves only as a predictive benchmark. The fourth and main model is a Random Forest regressor [25]. It averages many decision trees, each trained on a bootstrap sample of the training data. This lowers the variance compared with a single tree and still allows for nonlinear relationships and interactions between features.

The Random Forest uses the `RandomForestRegressor` from `scikit-learn` 1.6.1. The forest has 500 trees, bootstrap sampling, squared-error splitting and random seed 42. All other parameters keep their defaults: every feature is considered at each split, trees have no depth limit, a node needs at least two observations to be split, and a leaf may hold a single observation. Trees grown this deep will fit the training data almost perfectly. For that reason, out-of-bag predictions are kept (`oob_score=True`), which gives a held-out accuracy estimate inside the training period. Section 4.2 reports the training and out-of-bag scores next to the test score. Unlike Ridge, the Random Forest gets unscaled features, since tree-based models do not need standardized inputs. Both models are fitted on the 2016 to 2019 training set only. To check both settings on equal terms, Ridge and the Random Forest were validated on the same three expanding-year folds inside the training period (2016 to predict 2017, 2016–2017 to predict 2018, 2016–2018 to predict 2019), ranked by mean RMSE, with scaling refitted in each fold. The forest was compared with 12 settings

of maximum depth (none or 10), minimum leaf size (1, 3 or 5) and share of features per split (1.0 or 0.7). We decided beforehand to keep the default setting unless another one was more than 2% better. No observation from 2020 to 2023 is used for fitting or for choosing hyperparameters.

The Random Forest is also a randomized estimator. Each tree grows on a different bootstrap sample, so a different seed gives a different forest and slightly different predictions. One fit is really a single draw from a range of possible results. To see how wide that range is, the model is refitted 30 times with seeds 0 to 29 and nothing else changed. Section 4.2 reports the spread of the test metrics across these refits, and Section 4.5 gives the spread of the SHAP importance values. All other results in the paper come from a single reference fit with seed 42.

3.6. Evaluation Metrics

Model performance is measured on the held-out test set, which covers 2020 and 2021. Four regression metrics are used: root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE) and R^2. Each one picks up something the others miss, so all four are reported side by side.

RMSE is the square root of the mean squared error:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4}$$

where y_i is the observed gas consumption and $\hat{y}_i$ is the predicted gas consumption. RMSE has the same unit as the target variable, cubic meters per dwelling per year. Since the errors are squared first, a few large misses can push RMSE up a lot.

MAE is the mean of the absolute errors:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{5}$$

It is also expressed in cubic meters per dwelling per year, and it is the easiest of the four to read. It simply gives the typical size of a prediction error.

MAPE expresses the absolute error as a percentage of the observed value:

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{6}$$

Because it has no unit, MAPE allows comparison with studies that use other units or have different average consumption levels. The metric becomes unreliable when observed values are close to zero. That does not happen here, since every observed gas consumption value is positive and far above zero.

R^2 is the share of the variation in observed gas consumption that the model explains:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{7}$$

where $\bar{y}$ is the mean observed gas consumption in the split being evaluated. A higher value means the model captures more of the differences between municipality-year observations.

Confidence intervals come from a cluster bootstrap over municipalities: each of $B = 2{,}000$ replicates draws the 356 test municipalities with replacement and keeps all their test rows. The models are not refitted; the fixed predictions are rescored for all four models, and the 95% interval is the 2.5th to 97.5th percentile.

All four metrics are computed for the mean baseline, the persistence baseline, Ridge regression and the Random Forest. The mean baseline uses no predictors at all, so it shows how large the error is without any information. The persistence baseline only looks at a municipality's own past gas use. Ridge regression is a regularized linear model trained on the same features as the Random Forest. The Random Forest only counts as useful if it clearly beats both benchmarks.

We also report prediction errors in carbon dioxide equivalent. To convert them, we use the Dutch standard emission factor for G-gas: 1.779 kg CO_2-eq per Nm^3 [39]. This is one national factor for combustion only (tank-to-wheel). Upstream emissions are not included. It also ignores that gas composition differs a little between regions and years, for example when green gas is mixed in. But the conversion is linear and we only use it for reporting, so it does not change any ranking or comparison in the paper. If the factor is wrong, every emission figure is wrong by the same percentage. With a factor 5% lower, for example, 0.48 Mt would become 0.46 Mt. We use the same factor to express the differences between municipalities as emissions (Section 4.1).

National totals come from multiplying the mean gas use per dwelling by the number of dwellings in each municipality and adding up the results. To give a sense of how large the differences are, two convergence scenarios are computed. In the first, every municipality above the median brings its gas use per dwelling down to the median level. The second does the same with the 25th percentile as the target, and in both cases the result is the amount of gas use that would disappear. Both scenarios use the proxy per non-district-heated dwelling described above. Without this correction, municipalities with a lot of district heating would look efficient just because of how dwellings are counted.

The scenarios are accounting identities. They show how much gas use would go away under a simple assumption and say nothing about what could technically be saved. They are not expressed as a share of the 2030 target of the Dutch Climate Agreement [1], because that target is measured against a 2030 reference scenario, while the scenarios describe observed years.

3.7. SHAP Interpretation Framework

SHAP is used to explain the predictions of the Random Forest model. The aim is to find out which climate, building-stock and socioeconomic variables drive predicted municipal gas consumption the most. SHAP splits each prediction into parts, one per feature, and shows how much each part adds to or takes away from the result [4]. In this study the SHAP values have the same unit as the target variable: cubic meters of gas per dwelling per year. So a positive SHAP value raises the predicted gas use for a municipality-year, and a negative one lowers it.

The values are computed with `TreeExplainer` from the Python `shap` package. It was designed for tree-based models and runs fast on Random Forests [40]. We use `shap` 0.49.1 with its default tree-path-dependent setting, which handles feature dependence through the training data in each tree node and needs no background dataset. The explainer is applied to every observation in the 2020 to 2021 test set, which are the same observations used to evaluate model performance, and, as a check, to the 2016 to 2019 training rows.

For observation i, the Random Forest prediction breaks down as

$$\hat{y}_i = E[f(X)] + \sum_{j=1}^{p} \phi_{ij}, \tag{8}$$

where $\hat{y}_i$ is the predicted gas consumption, $E[f(X)]$ is the average model prediction, p is the number of predictors, and ϕ_{ij} is the SHAP value of feature j for observation i. Put

simply, every prediction starts at the average prediction, and each feature then moves it up or down by its own amount.

Global feature importance is the mean absolute SHAP value of each feature over the test set:

$$I_j = \frac{1}{n} \sum_{i=1}^{n} |\phi_{ij}|, \tag{9}$$

where I_j is the importance of feature j and n is the number of test observations. A feature gets a high score when it moves predictions a lot on average, in either direction.

I_j is an average over the test observations. That makes it an estimate, not a fixed property of the model, so Section 4.5 reports it with a confidence interval from the same cluster bootstrap. The SHAP values are computed once and each replicate recomputes I_j from the drawn rows. The ranking gets its own check: for each feature, the share of replicates in which it keeps the rank it has in the reference model. Seed variation is measured separately, by recomputing I_j for each of the 30 refits. The two most correlated pairs (income per resident with WOZ value, and single-family with owner-occupied share) are also ranked as groups, by summing their SHAP values per observation.

A SHAP beeswarm plot shows the direction and spread of the effects. Each point is one test observation for one feature, placed at its SHAP value and colored by the observed feature value. From this plot you can see which features matter most, and also whether high or low values push predicted gas use up.

These results describe how the Random Forest makes its predictions. They are not causal effects. Many municipal features are related to each other: floor area, the share of single-family housing, income and population density tend to move together. SHAP shows how the model uses these features. But it does not prove that changing one of them alone would change gas consumption by the same amount.

3.8. Use of Generative AI

Generative AI tools were used to help produce the figures and to support code development, source checks, and data interpretation. The authors verified all code and results and take full responsibility for the content of this paper.

4. Results

4.1. Descriptive Patterns in Gas Consumption

Figure 2 shows that average residential natural gas use varies a lot between Dutch municipalities. Across the 2016–2023 panel, `gas_m3` has a mean of 1,305 m^3 per dwelling per year and a median of 1,330 m^3. The middle half of the observations (the interquartile range) falls between 1,090 and 1,520 m^3. But the extremes are much further apart, from 310 m^3 at the low end to 2,550 m^3 at the high end. That is a large gap for figures measured per dwelling. The differences come from something other than some municipalities simply being bigger. We checked the extreme values against the source and keep them. The lowest values belong to municipalities where more than half of dwellings are on district heating, and the highest to wealthy municipalities with large detached homes. Refitting without the 32 training rows outside the 1st and 99th training percentiles (607 and 2,038 m^3) raises test RMSE from 112.5 to 124.4 m^3. The same municipalities return in the test years, and without their training rows the model cannot predict them well.

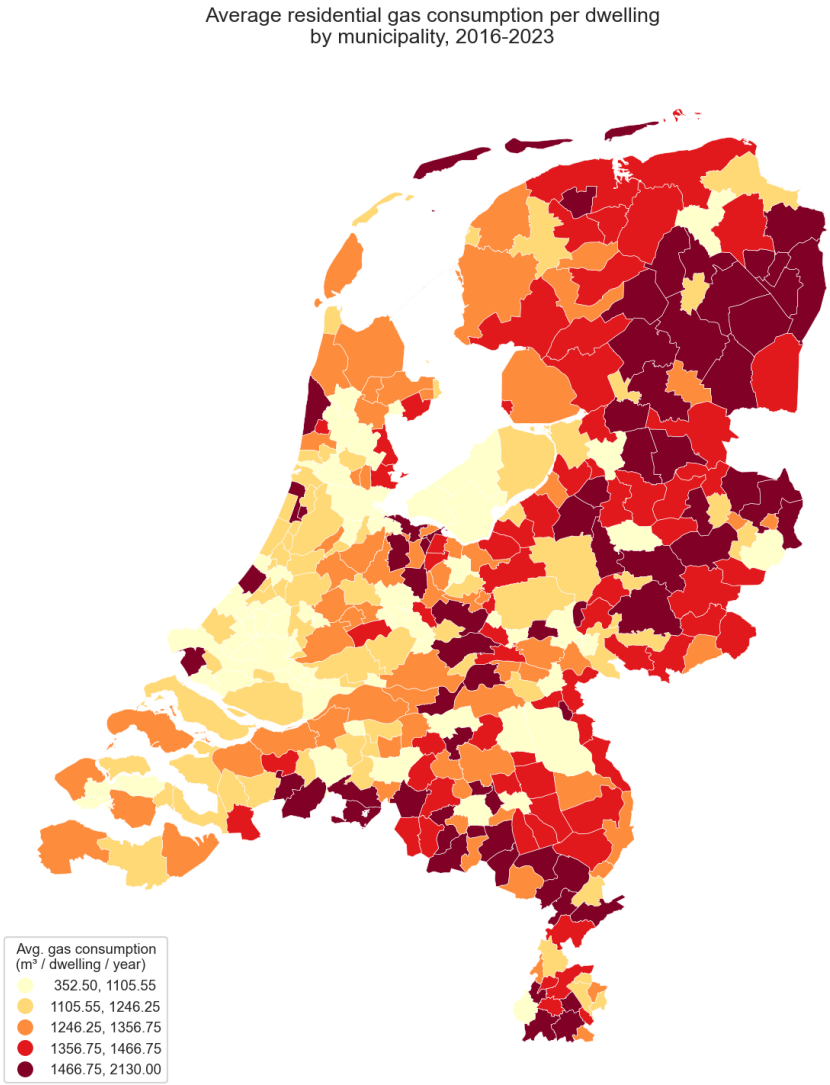


Figure 2. Average annual residential natural gas consumption per dwelling per municipality in the Netherlands, averaged over 2016–2023. Darker shading indicates higher consumption.

Average gas use is highest in small, less urban municipalities. Homes there tend to be large and are often single-family houses, and district heating is rare or absent. Rozendaal, Laren, Bloemendaal and Blaricum are examples, along with several rural municipalities in the north and east of the country. Many dwellings in these places are detached or semi-detached, with a larger average floor area.

Gas use is lowest in more urban or planned municipalities, where district heating is common or a bigger part of the housing stock is apartments. In the panel, Purmerend, Almere, Nieuwegein, Duiven, Utrecht, Rotterdam and Amsterdam are among the municipalities with the lowest consumption.

But much of this gap comes from how dwellings are counted, and says little about actual heating demand. Municipalities where more than 20% of dwellings have district heating use 454 to 662 m^3 less gas per dwelling than municipalities without district heating that have similar floor area and population density. If we divide the observed average by the share of dwellings not connected to a heat network, the difference drops to 15 to 48 m^3. The low average in these municipalities is therefore mostly the result of counting many dwellings that use little or no gas.

Figure 3 shows that gas use has gone down a lot over time. Mean municipal consumption dropped from 1,509 m^3 per dwelling in 2016 to 915 m^3 in 2023, a fall of about 39%. The

median tells almost the same story, going from 1,530 to 940 m^3. The path is not smooth, though. Consumption goes down every year until 2020 and then climbs back up in 2021. That year also had the highest heating demand in the panel, with a mean of 2,832 HDD, just above the 2,803 recorded in 2016. After 2021, consumption falls sharply. Weather explains only part of this drop. Both 2022 and 2023 were milder than the panel mean of 2,628 HDD, but the fall in consumption is too large to put down to climate alone. Other factors outside our data most likely played a role, especially the jump in energy prices during the 2022 energy crisis and the way households reacted to it. We don't measure these factors in this study, and we come back to them later.

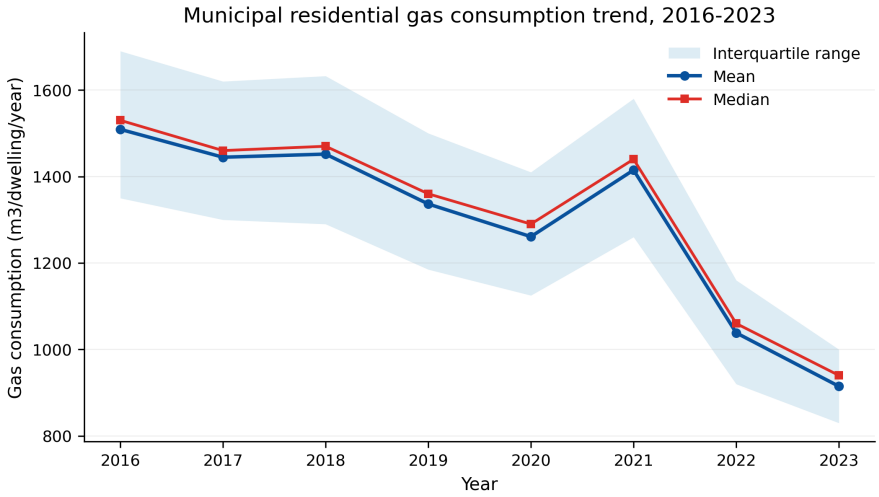


Figure 3. Annual trend in municipal residential natural gas consumption, 2016–2023. Lines show the mean and median across municipalities. The shaded band shows the interquartile range.

In absolute terms, municipalities also move closer together. The standard deviation falls from 263 m^3 in 2016 to 145 m^3 in 2023. But most of this drop just follows the lower mean, since the absolute spread usually shrinks when average consumption goes down. A relative measure is more useful here. The coefficient of variation (the standard deviation divided by the mean) stays at about 17.5% between 2016 and 2019, and by 2023 it has only fallen to 15.9%. So the relative gaps between municipalities hardly change, and the small narrowing we do see comes in the last two years. In 2023 some municipalities still use more than 1,400 m^3 per dwelling, while the lowest are close to 300 m^3. Consumption falls across the country over time, yet the differences between municipalities remain large.

These differences also matter in emission terms. Table 3 converts them to CO_2-eq for the two test years and for 2023. In 2020 and 2021, the gap between the 10th and 90th percentile municipality was on average 513 m^3 of gas per dwelling per year. That is 913 kg CO_2-eq. Using the proxy per non-district-heated dwelling, the gap drops to 450 m^3, or 800 kg CO_2-eq.

Across all dwellings, residential gas use caused about 16.7 Mt CO_2-eq per year. Suppose every municipality above the median had used gas at the median level. Emissions would then have been 0.48 Mt lower each year. Convergence to the 25th percentile would remove 1.28 Mt.

Policy cannot simply collect these numbers as savings. Part of the gap comes from climate and dwelling size, and policy has no control over either. In 2023 every value is lower, because the energy crisis cut gas use everywhere. But gas use above the median still added up to 0.28 Mt.

Table 3. Differences in residential gas use between municipalities, expressed in emissions with the factor of 1.779 kg CO_2-eq per Nm^3. Adjusted values divide gas use by the share of dwellings not connected to a heat network. Convergence scenarios give the national emissions above the median or 25th percentile municipality, using these adjusted values. The last column is the mean of 2020 and 2021.

Quantity	Unit	2020	2021	2023	Test mean
Per dwelling					
National mean per dwelling	t CO_2-eq	1.99	2.23	1.46	2.11
P10 to P90 gap, all dwellings	kg CO_2-eq	865	961	553	913
P10 to P90 gap, adjusted proxy	kg CO_2-eq	747	854	463	800
National					
Residential gas use	Mt CO_2-eq	15.68	17.79	11.77	16.73
Convergence to median	Mt CO_2-eq	0.44	0.51	0.28	0.48
Convergence to 25th percentile	Mt CO_2-eq	1.18	1.37	0.77	1.28

Figure 4 shows the Pearson correlations between the predictors and the target over the full panel. Many predictors are strongly correlated with each other. This is expected, since they describe related parts of the same local housing and population structure. The strongest pair is income per resident and average WOZ value ($r = 0.90$). Single-family housing share rises with owner-occupied share ($r = 0.82$) and falls as population density goes up ($r = -0.79$). It is also linked to average floor area ($r = 0.69$), which in turn is lower in denser municipalities ($r = -0.62$). The tenure variables form their own tight group, with owner-occupied share, social housing share and the share of single-person households all strongly correlated with each other. This affects interpretation in two ways. It is why the Ridge coefficients are used only as a predictive benchmark and are not interpreted substantively (Section 3.5). It also means SHAP attributions can be shared or shifted between correlated features, which is discussed in Section 3.7. With the target, however, no single predictor explains most of the variation on its own. The largest absolute correlation is 0.58, for heating degree days, which corresponds to about a third of the variance ($r^2 \approx 0.34$). This suggests that gas use depends on several features working together, with nonlinear effects and interactions, and it is the main reason for comparing the additive Ridge model with the Random Forest in Section 4.2.

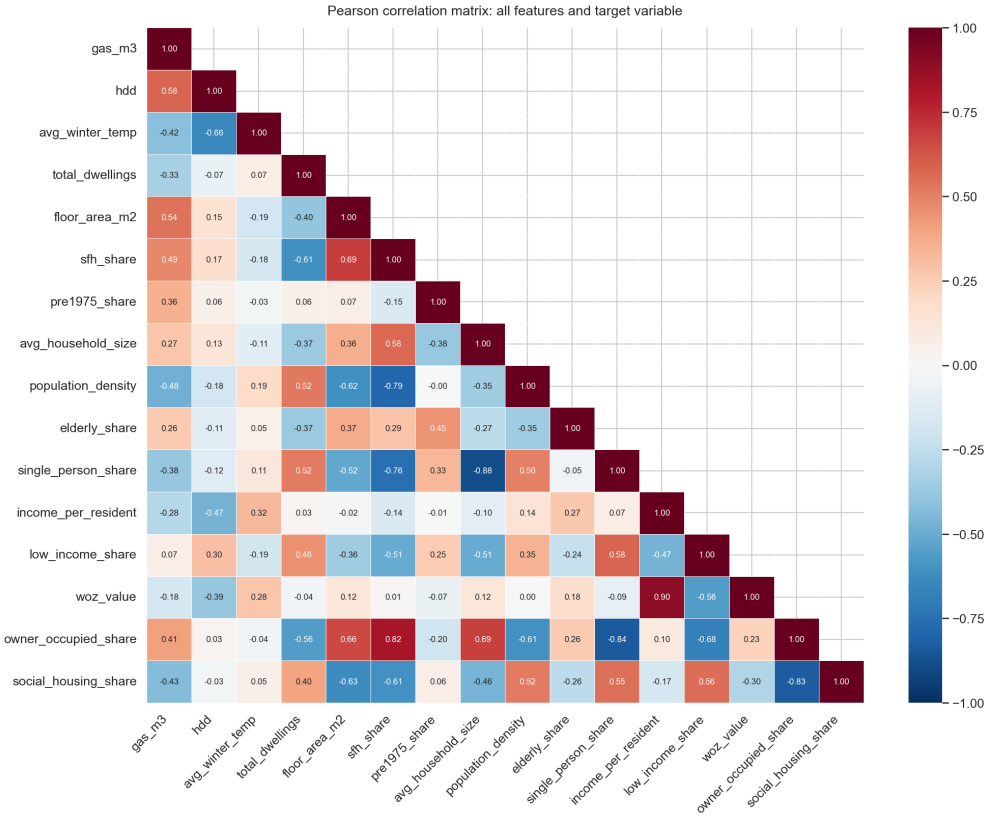


Figure 4. Pearson correlation matrix over the full 2016–2023 panel, covering the target (`gas_m3`), the 14 model predictors, and total dwellings. Color and value give the correlation coefficient.

These patterns shape the modelling approach in the next sections. The downward trend over time suggests that climate and other time-varying conditions matter. Structural municipal characteristics seem to matter as well, given how stable the spatial differences are. Building stock and urban density look especially relevant for explaining why some municipalities use much more gas per dwelling than others, and so does the presence of district heating.

4.2. Model Performance

Table 4 shows how the four models score on the four evaluation metrics. We report results for the 2016 to 2019 training set and for the 2020 to 2021 test set, which the models did not see during training. For the Random Forest, the table also gives the out-of-bag estimate. All models use the same time-based split, with 1,513 training observations from 2016 to 2019 and 707 test observations from 2020 to 2021. Together they cover 408 municipalities. And the two feature-based models use the same set of 14 structural predictors.

Table 4. Predictive performance on the 2016–2019 training set and the held-out 2020–2021 test set. RMSE and MAE are in m^3 per dwelling per year. The out-of-bag (OOB) row is a held-out estimate computed on the training rows and is defined only for the Random Forest. The persistence baseline has a test row only, because it repeats the 2019 value. Best test value in each column in bold.

Model	Split	RMSE	MAE	MAPE	R^2
Mean baseline	Train (2016–2019)	258.6	197.9	16.0%	0.00
	Test (2020–2021)	258.8	196.3	18.0%	−0.18
Persistence	Test (2020–2021)	**88.0**	**79.5**	**5.9%**	**0.86**
Ridge regression	Train (2016–2019)	112.4	82.2	6.6%	0.81
	Test (2020–2021)	138.2	108.9	9.1%	0.66
Random Forest	Train (2016–2019)	28.2	21.1	1.6%	0.99
	OOB (2016–2019)	76.4	57.4	4.3%	0.91
	Test (2020–2021)	112.5	92.6	7.6%	0.78

The most accurate model in Table 4 is persistence. Simply repeating each municipality's 2019 value gives an RMSE of 88.0 m^3 and a MAPE of 5.9%. That beats both feature-based models on every test metric. This makes sense, because past gas use already contains local detail that the 14 features can only approximate. So for a municipality we already know, the Random Forest is the weaker choice for predicting next year's consumption, and we do not claim otherwise. We use it to ask something else, namely how much of the difference between municipalities comes from their climate and building stock, together with their socioeconomic conditions. Persistence has nothing to say about this. It gives back the same number every year, and for a municipality with no consumption history it gives no number at all. The Random Forest can also be applied to municipalities it has not seen before, but with lower accuracy ($R^2 = 0.62$, see below).

Of the two feature-based models, the Random Forest is better on every test metric. It explains 78% of the variation in municipal gas consumption across the test set ($R^2 = 0.78$, 95% CI 0.73 to 0.81), and its predictions sit within 7.6% of observed consumption on average (MAPE, 95% CI 6.8% to 8.5%). The mean absolute error is 92.6 m^3 per dwelling per year, small next to the test-set mean of about 1,338 m^3, so a typical prediction is off by roughly 7%. And in emission terms that gap is 165 kg CO_2-eq per dwelling per year, against a municipal average footprint of 2.38 t.

Both predictive models beat the mean baseline by a wide margin. The baseline gives an R^2 of -0.18, so predicting one training-period average for every municipality-year does worse than the test-set variation alone would suggest. Average consumption in 2020 and 2021 sits below the 2016 to 2019 training mean, which makes a constant prediction biased from the start. Ridge regression closes most of the gap to the mean baseline, reaching an R^2 of 0.66 and a MAPE of 9.1% on the same feature set. The Random Forest goes further and cuts RMSE from 138.2 to 112.5 m^3, with MAPE falling from 9.1% to 7.6%. Part of the link between municipal characteristics and gas consumption is therefore nonlinear, or it runs through interactions between predictors that an additive Ridge model cannot pick up. Leave-one-out cross-validation selected $\alpha = 10$ for Ridge. On the expanding-year folds, Ridge selects a stronger penalty ($\alpha \approx 316$) but scores almost the same on the test set ($R^2 = 0.65$, MAPE 9.6%), so the comparison with the Random Forest does not depend on this choice. On the same folds, the best of the 12 forest settings lowered validation RMSE by only 0.7%, so the default was kept.

Every value in Table 4 comes from one fitted forest evaluated on one test set, so each is an estimate instead of an exact quantity. Table 5 repeats the test-set results with 95% confidence intervals, obtained from the cluster bootstrap over municipalities described in Section 3.6.

Table 5. Test-set performance with 95% confidence intervals from a cluster bootstrap over municipalities ($B = 2{,}000$ replicates, 356 municipalities drawn per replicate). Point estimates are those of Table 4. RMSE and MAE are in m^3 per dwelling per year.

Model	RMSE	MAE	MAPE	R^2
Mean baseline	258.8 [233.5, 285.0]	196.3 [180.6, 212.8]	18.0% [15.4, 21.1]	−0.18 [−0.26, −0.10]
Persistence	88.0 [82.9, 95.5]	79.5 [77.0, 82.5]	5.9% [5.8, 6.1]	0.86 [0.82, 0.89]
Ridge regression	138.2 [125.6, 152.9]	108.9 [101.6, 117.4]	9.1% [7.9, 10.7]	0.66 [0.61, 0.71]
Random Forest	112.5 [104.1, 121.5]	92.6 [87.1, 98.7]	7.6% [6.8, 8.5]	0.78 [0.73, 0.81]

The ordering of the models does not depend on the sample. For R^2 the Random Forest interval runs from 0.73 to 0.81 and the Ridge interval from 0.61 to 0.71, and the two do not overlap. RMSE, MAE and MAPE separate them in the same way. Every replicate is scored for all four models, so the models can also be compared within a replicate. There the Random Forest has the lower RMSE in 100% of the 2,000 replicates, by between 16 and 36 m^3 per dwelling per year. Refitting the forest with 30 different random seeds moves the test R^2 only between 0.777 and 0.788, and for every metric the spread over the seeds covers only 13% to 21% of the confidence interval, so the precision of these results comes from which municipalities end up in the sample and hardly at all from the fitting procedure. But the intervals are wide enough to matter. An R^2 of 0.78 is consistent with values from 0.73 to 0.81, and the MAPE of 7.6% could sit anywhere between 6.8% and 8.5%. So comparisons with other studies should quote the interval rather than the single figure.

Accuracy is clearly lower for municipalities the model has not seen before. When random groups of municipalities are held out, R^2 drops to 0.62, and with geographic blocks it drops to 0.58. Ridge does almost as well in both designs, and the gap between the two models is not significant. So a large part of the Random Forest's lead in Table 4 comes from municipalities it has already seen. Location on its own predicts poorly (a geography-only baseline has R^2 close to zero). Adding coordinates to the forest also hardly changes its accuracy or its SHAP ranking, which means the predictors are not simply standing in for region. The residuals are still spatially clustered, so the bootstrap intervals are likely somewhat too narrow.

Table 6 shows how much each feature group adds to the model. We refit the forest with the same settings on every combination of the three groups, and also on two versions that use only one climate variable each. The three building variables on their own get close to the full model. Without them, the test RMSE goes up by 25.3 m^3 (95% CI 9.6 to 41.3, municipality bootstrap). Without the socioeconomic variables it goes up by 4.0 m^3 (−4.1 to 11.8), and since this interval includes zero we cannot call it a clear difference. But removing the two climate variables makes the RMSE lower, in both the validation and the test years. It could be that climate changes mostly from year to year and much less between municipalities. The training period has only four years, so the forest can partly use climate as a label for the year, and this does not help for later years. The two single-climate versions also do not agree with each other: HDD alone does better on validation, while winter temperature alone does better on test. For this reason we keep the full model as specified. Our reading of the table is that building-stock characteristics do most of the work in the prediction and socioeconomic context adds a little. Year-to-year climate does not improve out-of-sample accuracy in this panel.

Table 6. Random Forest performance by feature group (C = climate, B = building stock, S = socioeconomic), with the reference settings. Validation RMSE is the mean over the three expanding-year folds within 2016–2019; the other metrics are on the 2020–2021 test set. RMSE and MAE are in m^3 per dwelling per year.

Features	n	Val. RMSE	RMSE	MAE	MAPE	R^2
C	2	310.7	251.1	185.5	16.5%	−0.11
B	3	102.3	118.8	97.3	7.5%	0.75
S	9	119.9	141.8	110.5	9.6%	0.65
C+B	5	121.0	116.5	95.0	7.5%	0.76
C+S	11	135.9	137.8	99.3	8.9%	0.67
B+S	12	96.1	109.0	90.0	7.3%	0.79
C+B+S (reference)	14	105.3	112.5	92.6	7.6%	0.78
HDD only + B+S	13	95.0	116.2	97.4	8.0%	0.76
Winter temperature only + B+S	13	118.7	100.8	81.5	6.8%	0.82

Figure 5 plots predicted against observed gas consumption for the Random Forest on the test set. The points stay close to the 45-degree line. That holds across the whole range, from under 600 to more than 2,000 m^3 per dwelling per year, so the model captures the wide differences between municipalities described in Section 4.1 and still gets the low and high ends roughly right. Section 4.3 looks at where the remaining errors come from.

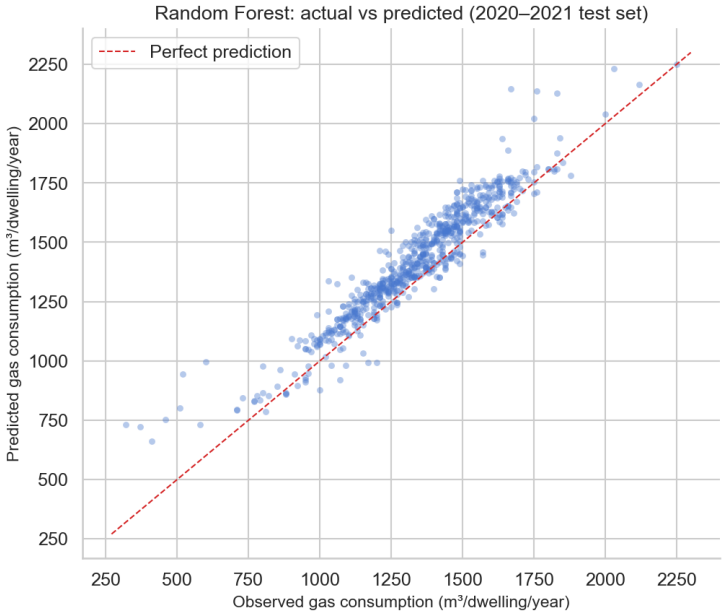


Figure 5. Random Forest predicted versus observed municipal gas consumption on the 2020–2021 test set. The dashed line marks perfect prediction.

Table 7 shows the results of the two district-heating checks. When the district-heating share is added as a feature, the Random Forest test RMSE falls from 112.5 to 103.8 m^3 per dwelling. But this is still worse than persistence, which reaches 88.0. In the SHAP ranking, the share comes in third, and pre-1975 housing falls from fourth place to seventh. With the adjusted target, the top three do not change (floor area, HDD and population density), while winter temperature moves into fourth place, ahead of pre-1975 housing. So the main drivers hold.

Table 7. Random Forest test performance (2020–2021) and the four most important SHAP features for the reference model and the two district-heating checks. RMSE and MAE are in m^3 per dwelling, except for the adjusted target, where they are in m^3 per non-district-heated dwelling. Scores for the adjusted target are therefore not directly comparable with the other two rows.

Variant	RMSE	MAE	MAPE	R^2	Top four SHAP features
Reference (14 features)	112.5	92.6	7.6%	0.78	Floor area, HDD, population density, pre-1975 share
Plus district-heating share	103.8	87.4	6.7%	0.81	Floor area, HDD, district heating, population density
Adjusted target (14 features)	104.9	88.2	6.6%	0.72	Floor area, HDD, population density, winter temperature

4.3. Prediction Error Patterns

The Random Forest gets the differences between municipalities right, but in the test years it puts the overall level a bit too high. Across the 707 test observations, the mean observed gas use is 1,337.6 m^3 per dwelling per year. The model predicts 1,420.8 m^3. That is 83.2 m^3 too much on average, about 6% above the observed mean.

Figure 6 plots the residuals, the observed value minus the predicted value, against the predicted value. Almost every point sits below zero, so the model predicts too high for most municipalities. The average error of 83.2 m^3 is about 90% of the mean absolute error of 92.6 m^3, which means the errors mostly point the same way instead of cancelling out. The overprediction shows up across the whole range of consumption. But the ranking of the municipalities still holds up, with a test-set R^2 of 0.78.

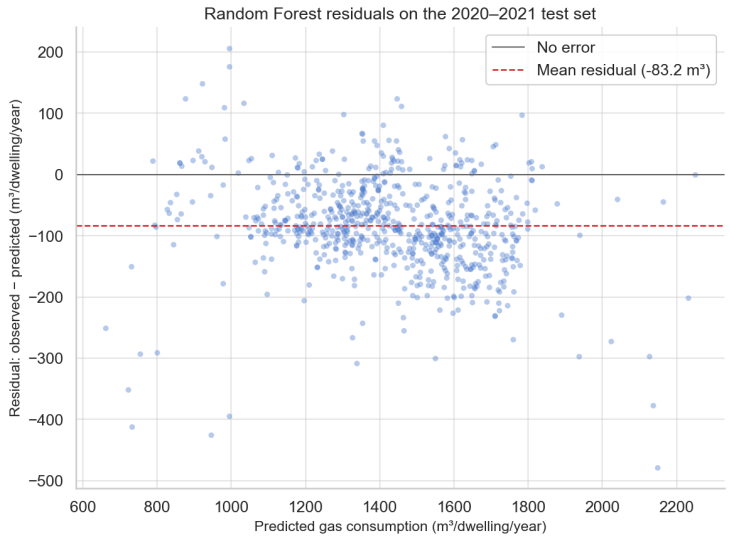


Figure 6. Random Forest residuals (observed minus predicted) on the 2020–2021 test set, plotted against predicted consumption. The solid line marks zero error and the dashed line the mean residual of −83.2 m^3. Most points lie below zero, showing a small and fairly even overprediction.

This shift comes from time. It is not caused by differences between municipalities. The model is trained on 2016 to 2019 and used for 2020 to 2021. For the same weather and building characteristics, dwellings used a little less gas in the test years than in the training years, for reasons the feature set does not capture. The model learned the relationship from the training period, so it predicts this small amount too high. Such drift is not new: a backtest trained on 2016 to 2018 overpredicts 2019 by 123.0 m^3. The overprediction is about the same in both test years, −85.0 m^3 in 2020 and −81.3 m^3 in 2021, even though average use was lower in 2020 and higher in 2021. So the model follows the change from year to year well, and the error is a steady offset in level. The same effect gives the mean

baseline a negative R^2 in Section 4.2, but it is much smaller here, because the Random Forest captures the differences between municipalities through its predictors. And this same shift is what separates the out-of-bag R^2 of 0.91 from the test R^2 of 0.78. When we add the year as a predictor, fitted on the training years only, the mean residual is cut in half, to -39.3 m^3 (R^2 0.85). This happens because the forest then uses the 2019 level. But the order of municipalities within each year stays almost the same: the Spearman correlations are 0.97 in 2020 and 0.95 in 2021, compared with 0.97 and 0.94 for the reference model. We keep the reference model. If we picked a variant after seeing these test scores, we would be selecting on the test set.

This bias is a small problem for what the study is trying to do. The aim is to explain how municipal characteristics relate to gas consumption, not to forecast exact future values. It is also small next to the spread between municipalities, which runs from below 600 to above 2,000 m^3 per dwelling per year. And because the bias shifts nearly all predictions by about the same amount, it leaves the differences between municipalities, which are what the SHAP analysis explains, largely intact. The model would still need adjusting before anyone used it to predict actual values for years after the training period.

4.4. Out-of-Distribution Evaluation

The errors described so far come from the in-regime test set, where market conditions look much like they did during training. The model is also run on the 685 held-out observations from 2022 and 2023, the years of the European energy crisis. This is a harder test for the structural features. These observations are never used for fitting or model selection (Section 3.5), so they fall outside the training distribution. Table 8 puts the two windows side by side.

Table 8. Random Forest performance on the in-regime test set (2020–2021) and the out-of-distribution stress-test set (2022–2023). RMSE, MAE, and the mean residual are in m^3 per dwelling per year. The mean residual is observed minus predicted.

Window	n	RMSE	MAE	MAPE	R^2	Spearman	Mean residual
In-regime (2020–2021)	707	112.5	92.6	7.6%	0.78	0.96	−83.2
OOD (2022–2023)	685	449.4	424.6	45.1%	−5.83	0.77	−424.6

On the OOD window the accuracy falls apart. RMSE climbs from 112.5 to 449.4 m^3, almost four times higher. MAPE goes from 7.6% to 45.1%, close to a sixfold jump. The R^2 drops from 0.78 to -5.83. A negative R^2 means the model does worse than simply predicting the mean of the 2022 to 2023 observations, so at this point it carries no useful signal. And the error keeps growing year by year. In 2022 the R^2 is -3.34; by 2023 it has fallen to -11.98, with MAPE rising from 34.7% to 55.7% over the same period.

The model misses a sharp drop in consumption. Observed use falls from a mean of 1,337.6 m^3 per dwelling in the in-regime window to 977.1 m^3 in the OOD window, about 27% lower. But the mean prediction barely moves, from 1,420.8 to 1,401.6 m^3. The features driving it are structural ones, like floor area, building age and population density, and those change little from one year to the next. So the mean residual grows from -83.2 m^3 in-regime (-6.2% of observed) to -424.6 m^3 (-43.5%), roughly five times larger. The model expects a normal year.

Compared with the last training year, mean use fell by 298 m^3 in 2022 and by 422 m^3 in 2023. Before 2022, the biggest change from one year to the next was 115 m^3 (in 2019), so these drops are three to four times larger. The municipalities also became more similar to each other, and this smaller spread pushes R^2 even further below zero. So the failure is the same level drift we saw in earlier years, only much bigger. High prices and conservation

campaigns are the most likely reason. But we do not measure either of them, so we call this a failure during the crisis period and do not claim that the crisis caused it. We tried two time corrections, and neither one fixes the problem reliably. With the year predictor, the OOD mean residual is still -409.6 m^3. The linear trend is extrapolated from only four noisy training years. It happens to reduce the residual to -109.6 m^3, but it then underpredicts 2020 to 2021 by 99.6 m^3.

The ranking of municipalities holds up much better than the level. On the pooled OOD window, the Spearman rank correlation between observed and predicted consumption falls only from 0.96 to 0.77, and it stays higher inside each year, 0.92 in 2022 and 0.85 in 2023. So the model still shows which municipalities use more gas than others, even when it puts the level in the wrong place. Section 3.5 led us to expect this. Structural municipal features carry the differences between municipalities. But nothing in them can see a national demand shock coming. The rank correlation does decline from 2022 to 2023. For the 339 municipalities observed in both years it falls from 0.92 to 0.85, a difference of -0.08 (95% CI -0.12 to -0.05, paired municipality bootstrap, 2,000 resamples). So in the crisis years the gaps between municipalities also changed, not only the overall level, although the data do not show why. That is a good reason to keep the main analysis in the pre-crisis period, and to add price and policy variables in future work (Section 5).

4.5. Global Feature Importance

SHAP values come from TreeExplainer, applied to the 707 held-out observations from 2020 to 2021. This is the same test set used for the performance evaluation in Section 4.2. Values are given in cubic meters per dwelling per year, so the importance scores carry the same unit as the target variable. Global importance I_j is the mean absolute SHAP value per feature. The full ranking is in Table 9. Figure 7 plots how the SHAP values spread across the test observations.

Table 9. Global feature importance for the Random Forest model, measured as the mean absolute SHAP value over the 707 test observations. Values are in m^3 per dwelling per year.

Rank	Feature	Mean |SHAP| [95% CI]	Held	Ranks	Category
1	`floor_area_m2`	125.6 [117.8, 133.6]	100.0%	1	Building
2	`hdd`	53.4 [51.6, 55.2]	100.0%	2	Climate
3	`population_density`	43.8 [41.1, 46.1]	100.0%	3	Socioeconomic
4	`pre1975_share`	25.2 [20.2, 30.5]	99.8%	4	Building
5	`avg_winter_temp`	18.1 [17.5, 18.7]	82.0%	5–6	Climate
6	`income_per_resident`	16.4 [13.2, 20.2]	82.0%	5–6	Socioeconomic
7	`woz_value`	12.0 [11.1, 12.9]	89.0%	7–8	Socioeconomic
8	`social_housing_share`	11.2 [10.4, 11.9]	78.7%	7–9	Socioeconomic
9	`sfh_share`	10.5 [9.6, 11.3]	79.3%	8–10	Building
10	`elderly_share`	9.6 [8.8, 10.5]	90.5%	9–10	Socioeconomic
11	`single_person_share`	7.1 [6.5, 7.6]	100.0%	11	Socioeconomic
12	`low_income_share`	2.5 [2.3, 2.8]	100.0%	12	Socioeconomic
13	`owner_occupied_share`	1.4 [1.3, 1.6]	100.0%	13	Socioeconomic
14	`avg_household_size`	0.4 [0.4, 0.5]	100.0%	14	Socioeconomic

Average floor area does more work in the fitted Random Forest than any other predictor. Its mean absolute SHAP value is 125.6 m^3 per dwelling per year (95% CI 117.8 to 133.6), which is over a third of the total attribution across all features. Larger dwellings need more energy to heat, so the model is picking up something physically sensible. But the values fall away quickly after that. Heating degree days come second at 53.4 m^3 (51.6 to 55.2), then population density at 43.8 m^3 (41.1 to 46.1). The pre-1975 housing share comes fourth at 25.2 m^3 (20.2 to 30.5). Together these four make up 73.6% of the total mean absolute SHAP value (72.3% to 74.7%), so most of what the model explains runs through a small group of variables.

Three checks support this ranking. Across the 30 seed refits, the first seven ranks never change. On the training rows, the same four features come first (Spearman correlation with the test ranking 0.98). With the correlated pairs grouped, the first four places are unchanged; income and WOZ value together come fifth (23.4 m^3, 95% CI 19.4 to 27.9), close to `pre1975_share`.

All three feature categories appear in the ranking, and no single source takes over. The top four predictors are two building characteristics (`floor_area_m2`, `pre1975_share`), one climate variable (`hdd`) and one socioeconomic variable (`population_density`). And a second climate feature (`avg_winter_temp`) together with a second socioeconomic one (`income_per_resident`) comes right after. This shows that the model uses all three groups, but not that each one improves accuracy. The feature-group comparison in Table 6 shows that most of the accuracy comes from the building-stock variables.

These attributions show how the fitted Random Forest spreads credit across the predictors. They are not causal effects. Several municipal characteristics move together, for example floor area, single-family housing share, income per resident and WOZ property value, so credit for one of them can shift to a related predictor. The ranking is best read as a description of how the model used the available features. It does not isolate what each variable contributes on its own.

The rest of the features matter much less. From `avg_winter_temp` down, every feature has a mean absolute SHAP value below 19 m^3 per dwelling per year, far under the leading predictors, and average household size sits at 0.4 m^3, so the model barely uses it. But once floor area, climate, building age, and population density are in the model, several of the socioeconomic variables have little left to explain. That fits the correlations between municipal characteristics described above and in Section 3.7.

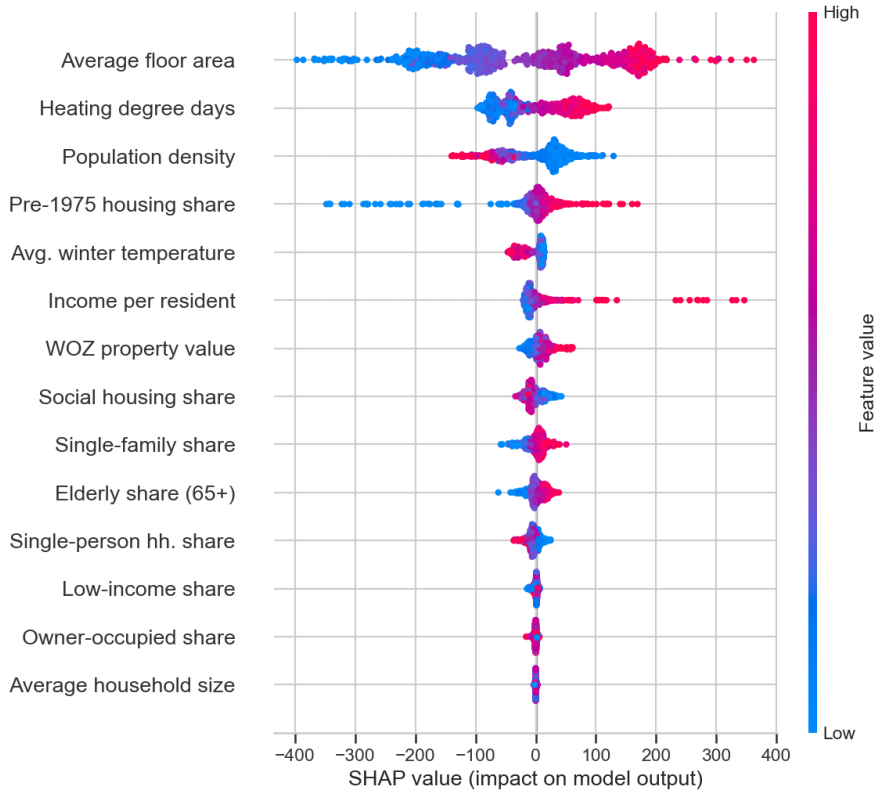


Figure 7. SHAP beeswarm plot for the Random Forest model on the 2020–2021 test set. Features are ordered top to bottom by mean absolute SHAP value. Each point is one municipality-year observation; horizontal position is the SHAP value in m^3 per dwelling per year, and color encodes the feature value from low (blue) to high (red).

Figure 7 keeps the same ordering and also shows the direction and spread of each effect. Average floor area and heating degree days have the widest horizontal spread. That fits their place at the top of Table 9. Section 4.6 looks at the direction of these effects.

4.6. Direction of Feature Effects

The beeswarm plot in Figure 7 shows how strong each feature effect is and which way it points. For the top predictors, low and high feature values sit on opposite sides of zero. So these features act in a consistent direction. Low values push the prediction one way, and high values do the opposite. Which side is which depends on the feature.

The main building and climate features point in the expected direction. A higher average floor area pushes predicted gas use up: municipalities with larger dwellings get positive SHAP values, and the ones with smaller dwellings sit on the negative side. This is also the widest spread in the plot, which fits with floor area being the most important feature in Table 9. Heating degree days work the same way. Colder municipality-years, with a higher `hdd`, push the prediction up, and milder ones pull it back down. But average winter temperature runs the other way. Warmer winters lower predicted gas use. The share of pre-1975 housing also raises predicted consumption, which supports reading this variable as a proxy for older and less insulated dwellings.

Figure 8 shows how the contribution of the four leading features changes with their value. In all four cases the contribution moves in one direction only, but never in a straight line. Each feature has a range where the slope changes quickly. For floor area, the contribution climbs fast between about 100 and 150 m^2, going from around -200 to $+120$ m^3. After that it flattens out at about $+170$ to $+200$ m^3, so a larger dwelling adds very little to the prediction once it is past this size. The `hdd` contribution stays at about -40 to -60 m^3 below 2,650. Then it rises steeply between 2,650 and 2,800. This step mostly splits the two test years, because 2020 was mild (`hdd` 2,192 to 2,732) and 2021 was cold (2,548 to 3,043). Some 2020 values are also lower than 2,345, the lowest value in the training data. The forest cannot extrapolate there, and the contribution stays flat. Population density adds about 20 to 40 m^3 below 700 inhabitants per km^2. It turns negative between 700 and 1,000 and then keeps falling more slowly, reaching about -90 m^3 above 3,000. The pre-1975 share rises gently above 0.3. But below about 0.27 it drops steeply, and this drop is based on only 31 of the 707 test rows. It should be read carefully.

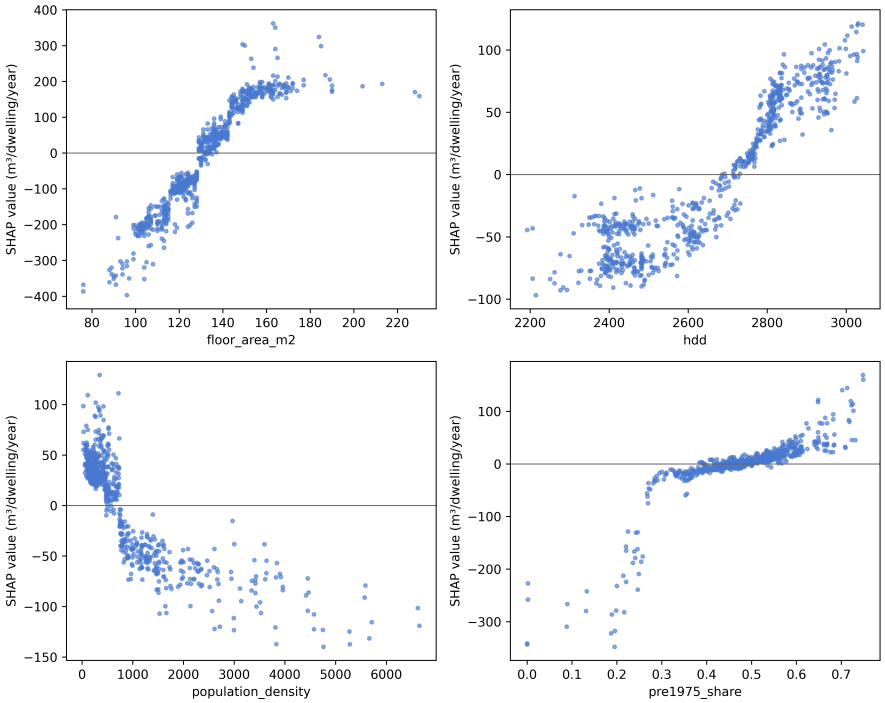


Figure 8. SHAP dependence plots for the four leading features of the Random Forest model on the 2020–2021 test set. Each point is one municipality-year observation. The horizontal axis is the feature value (floor area in m^2, heating degree days, inhabitants per km^2, and share of dwellings built before 1975); the vertical axis is its SHAP value in m^3 per dwelling per year.

The socioeconomic and urban-form features mostly split compact urban municipalities from the more spacious rural ones. Population density pushes the prediction down. Dense municipalities get negative SHAP values, and the sign flips once density drops off. But a larger share of single-family housing works the other way and raises the prediction, while municipalities with more social housing or more single-person households are pulled back down. None of this is surprising once you think about the housing stock behind it. Flats in compact urban areas usually burn less gas per dwelling than the bigger homes you find in low-density areas.

The affluence variables get positive SHAP values, but the story behind them is probably about housing rather than money. Income per resident and WOZ property value both push predicted gas use up. Wealthier municipalities tend to have larger dwellings and a bigger share of detached and semi-detached homes, so the income link is indirect. Elderly share also pushes the prediction up. Older residents spend more hours at home and usually keep the heating a bit higher. Low-income share does almost nothing, and what little it does is weak. Owner-occupied share has a small effect as well. And average household size is the variable the model barely touches.

These directions show how the fitted Random Forest uses the predictors. They are not causal effects. Several predictors move together, above all floor area, the share of single-family housing, income per resident, WOZ property value and population density. Because of that overlap, one underlying housing signal can end up split across several features, and the importance that lands on any single one of them depends partly on which correlated variable the trees happened to pick first. The direction is still informative. But the exact size attached to a single feature should not carry much weight.

5. Discussion

5.1. Interpretation of Results

The results show that municipal characteristics can predict annual residential gas consumption with useful accuracy, without any lagged consumption variable. The Random Forest reaches a MAPE of 7.6% (95% CI 6.8% to 8.5%) and an R^2 of 0.78 (0.73 to 0.81) on the 2020 to 2021 test set. Persistence is more accurate still, so the value of the model lies elsewhere. Each prediction is built from climate, dwelling stock and population, which makes it possible to ask which characteristics are associated with higher or lower gas use.

Ridge regression and the Random Forest use the same 14 predictors, yet the forest lowers RMSE from 138.2 to 112.5 m^3 (about 19%) and raises R^2 from 0.66 to 0.78. Part of the relationship is therefore nonlinear or runs through interactions that an additive model cannot represent. This advantage holds only for later years of municipalities seen in training. When whole municipalities are held out, the two models perform about equally (Section 4.2). Much of the forest's gain therefore seems to come from fitting the particular combinations of characteristics of known municipalities, not from structure that transfers to new ones.

The SHAP results show which variables the model relies on. Average floor area is the strongest predictor, its mean absolute SHAP value of 125.6 m^3 per dwelling per year is more than twice that of heating degree days. This makes sense: larger dwellings generally have more heated space and more surface through which heat can be lost. Floor area also correlates with single-family share and density, though, so part of its attribution likely stands in for those. Heating degree days are second, but the feature-group comparison shows that climate does not improve accuracy for later years. Population density ranks third and has a negative effect, which suggests that compact urban municipalities tend to use less gas per dwelling. The pre-1975 housing share raises predicted consumption. The four leading features together account for 73.6% of the total attribution, with a confidence interval of 72.3% to 74.7%.

This paper aimed to bring climate, building-stock and socioeconomic data together in one model, and the final model does use all three. But the three groups do not help equally, as the feature-group comparison shows. Most of the predictive accuracy comes from the building-stock variables. The socioeconomic variables add only a little, and this gain is not statistically significant. The climate variables do not improve predictions for later years. It cannot fully separate what each group does because building and socioeconomic variables are strongly correlated. It does show that most of what the model can predict about differences between municipalities comes through the building stock. SHAP then makes it possible to look inside the fitted forest. Its predictors can be ranked, and their effects can be checked against what we would expect physically.

In practice, the model works as a screening tool. What the model adds is a number: the expected level of gas use for each municipality, based on open data. When a municipality is left out of training, its prediction shows how much gas we would expect from municipalities with a similar building stock and climate. If a municipality is far above that level, it uses more gas than we would expect from its characteristics. That is a reason to look closer locally, for example at heating systems or renovation history, or at the quality of the data itself. But accuracy for unseen municipalities is lower (R^2 of 0.62), so only large deviations should be read this way. The model also cannot say whether insulation programs or heat pump subsidies would reduce gas use in a given place. SHAP values describe the fitted model. They do not show the effect of an intervention (Section 3.7).

The results link back to the Sustainable Development Goals from the Introduction in three ways. The first is energy efficiency (SDG target 7.3). Gas use per dwelling tells us something about efficiency, but it does not measure it directly, because it also depends

on the climate, and on how big the dwelling is and how it is heated. The second link is planning (SDG target 11.3): the results are made for the same planning unit that the Dutch Climate Agreement holds responsible, and they use open data only. The third is emissions. When gas use is converted to emissions, the results connect to SDG target 13.2. In the test years, observed gas use above the municipal median adds up to about 0.48 Mt CO_2-eq per year. Above the 25th percentile it is 1.28 Mt (Table 3). But policy can only close part of this gap, since some of it comes from climate and dwelling size.

5.2. Comparison with State of the Art

The closest municipal-scale comparison is Gaweł and Paliński [12], who forecast household natural gas consumption for Polish municipalities and found that global models performed better than local models. Their best municipal-level MAPE was 7.3%. The Random Forest in this study reaches a similar level of accuracy, with a MAPE of 7.6% on the 2020 to 2021 test set. The numbers are not directly comparable because the countries, input variables, and target definitions differ. The difference is mainly in the feature design and interpration aim. Gaweł and Paliński [12] used variables such as lagged consumption, population, dwelling counts, and gas access. This is useful for forecasting, but it gives less insight into the reasons why one municipality uses more as than another. Still, both studies support the use of a global model for this problem, especially when the goal is to learn shared patterns across many municipalities.

At smaller spatial scales, Jafarigorzin et al. [8] and Colverd et al. [9] report strong performance. These studies are important because they show that building and demographic variables are highly informative for local energy demand. However, their settings differ from this study. Jafarigorzin et al. [8] found that lagged consumption became the strongest predictor when it was included. When removed, building surface area and construction year became more important. Colverd et al. [9] also found that building count and floor area were important, with temperature as a secondary predictor. The SHAP results show a similar pattern at the municipal scale.

So predictive accuracy is not the main contribution of this paper. What the paper does show is which municipal characteristics go together with higher or lower observed annual gas use, based on open data and SHAP. In the feature-group comparison, most of the predictive accuracy comes from the building-stock variables. Dutch planning tools such as the Referentieverbruik method [13] estimate demand bottom-up from dwelling characteristics, and the local differences that remain end up in a correction factor. But this study works top-down. It tests how well open data can explain the observed differences between municipalities, and in that way it adds to the existing tools.

5.3. Limitations

Several limitations should be considered when interpreting the results. First, the SHAP analysis explains the fitted Random Forest model, not causal effects. A positive SHAP value for a feature means that the model uses that feature to raise the prediction for specific observations. It does not prove that changing that feature alone would reduce or increase gas consumption by the same amount. This is especially important because many municipal characteristics are correlated. Floor area, single-family housing share, income, WOZ value and population density are all related parts of the local housing structure. A different model could also show other results with these correlated variables. The exact attribution assigned to one variable should therefore be read with care.

Second, the main evaluation only tests later years for municipalities the model has already seen. But the screening use described above needs the model to work on municipalities that were left out of training. There, accuracy is clearly lower: R^2 is 0.62 for

random groups and 0.58 for geographic blocks, and the forest does no better than Ridge. The residuals are also spatially autocorrelated. This means neighbouring municipalities share part of their information, so the reported intervals are somewhat too narrow. We did not test transfer to another country.

Third, the analysis uses municipal averages. The scale is fine for municipal heat-transition planning, but it does not show variation within municipalities. A municipality can contain dense apartment neighborhoods, older village centers, and newer low-density suburbs at the same time. The model cannot identify these within-municipality patterns. For local implementation, municipal-level results should therefore be combined with neighborhood or dwelling-level data where available.

Fourth, gas consumption is one energy carrier, not the total heat demand of a dwelling. In hybrid heating systems, a heat pump covers most of the load and the gas boiler mainly covers cold spells and peaks. Their gas use therefore understates heat demand, and the split between gas and electricity also shifts with energy prices, as in 2022. No municipal data on heat pumps were available for the study years, so heating-system type is an omitted variable. It may explain part of the municipal differences and part of the overprediction in later years.

Finally, only one model family is examined in depth. The study covers a mean baseline, a persistence baseline, Ridge regression, and a Random Forest with a fixed configuration. Gradient boosting methods such as XGBoost, LightGBM, and CatBoost are not tested, and the forest was only checked against a small grid of 12 settings. This was a deliberate choice, because the aim of this paper is interpretation rather than model optimization. The feature groups were also compared with fixed settings, not tuned separately. A systematic comparison of model families on this panel is left for future work.

6. Conclusions

This paper makes four contributions. (1) It merges four Dutch open datasets into a municipality-year panel of 2,905 observations covering 408 municipalities from 2016 to 2023. (2) It estimates annual gas use per dwelling from open structural data, without past consumption, and compares the model with simple baselines and across feature groups. (3) It uses SHAP to show which municipal characteristics the model relies on to explain differences in gas use between municipalities. (4) It tests the model on the 2022 to 2023 energy crisis.

The best model on the 2020 to 2021 test set was the persistence baseline, which simply repeats the last observed value of each municipality (R^2 0.86, MAPE 5.9%). The Random Forest came second, with an R^2 of 0.78 (95% CI 0.73 to 0.81) and a MAPE of 7.6%. Ridge regression did worse (0.66 and 9.1%). So the forest is more useful for explaining gas use than for predicting it. It estimates gas use from structural characteristics, which means it can also be applied to municipalities that have no consumption history. But for municipalities left out of training, R^2 dropped to 0.62 and the forest did no better than Ridge. Most of the accuracy came from the building-stock variables. Socioeconomic variables added little, and climate variables did not improve prediction for later years. The four strongest predictors were average floor area, heating degree days, population density and the share of dwellings built before 1975. Together they held 73.6% of the SHAP importance. Once district heating was included, part of the effect of older housing shifted to the district-heating share. In the crisis years the model could not predict the level of gas use (R^2 -5.83). The ranking of municipalities held up much better (Spearman 0.92 in 2022 and 0.85 in 2023), although it clearly got weaker from one year to the next. In the test years, gas use above the municipal median added up to 0.48 Mt CO_2-eq per year. Policy can only reach part of this gap, because some of it comes from climate and dwelling size.

The main limitations are that SHAP attributions are not causal and are shared between correlated predictors, that accuracy is lower for unseen municipalities and the residuals remain spatially autocorrelated, that municipal averages hide differences within municipalities, that heating-system type could not be included, and that only one model family was studied in depth. Future work should: (1) add annual gas price and policy indicators to improve level accuracy; (2) use spatial models and test transfer to other countries; (3) repeat the analysis at neighborhood level, where Dutch heat transition plans are actually made; and (4) compare gradient boosting methods against the Random Forest on the same panel.

For Dutch municipalities, which carry most of the responsibility for the heat transition, the model is best used as a screening tool. It does not show which measures would reduce gas use, but it does show from open data which characteristics go together with high gas use. It also gives each municipality an expected level of gas use based on its structure, so that municipalities using clearly more than expected can be identified for local follow-up.

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